

## Ex. No: 13 File Organization Techniques

### PROGRAM A: SEQUENTIAL FILE ORGANIZATION

#### C PROGRAM

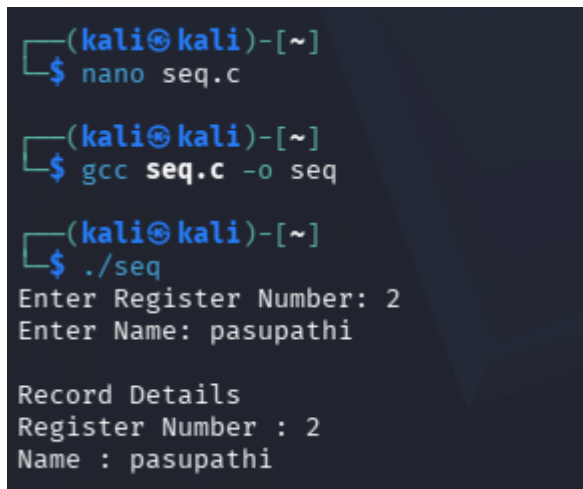
```
#include <stdio.h>

struct student
{
    int regno;
    char name[20];
};

int main()
{
    FILE *fp;
    struct student s;
    fp = fopen("student.dat", "w");
    File Organization Techniques
    printf("Enter Register Number: ");
    scanf("%d", &s.regno);
    printf("Enter Name: ");
    scanf("%s", s.name);
    fprintf(fp, "%d %s\n", s.regno, s.name);
    fclose(fp);
    fp = fopen("student.dat", "r");
    fscanf(fp, "%d %s", &s.regno, s.name);
    printf("\nRecord Details\n");
    printf("Register Number : %d\n", s.regno);
```

```
printf("Name : %s\n", s.name);  
fclose(fp);  
return 0;  
}
```

Output:



```
(kali㉿kali)-[~]  
$ nano seq.c  
  
(kali㉿kali)-[~]  
$ gcc seq.c -o seq  
  
(kali㉿kali)-[~]  
$ ./seq  
Enter Register Number: 2  
Enter Name: pasupathi  
  
Record Details  
Register Number : 2  
Name : pasupathi
```

SHELL SCRIPT

```
#!/bin/bash  
  
echo "Enter Register Number:"  
  
read regno  
  
echo "Enter Name:"  
  
read name  
  
echo "$regno $name" > student.txt  
  
echo "Contents of File"  
  
cat student.txt
```

Output:

```
(kali㉿kali)-[~]  
$ bash lab13-1.sh  
Enter Register Number:  
22  
Enter Name:  
kali  
Contents of File  
22 kali
```

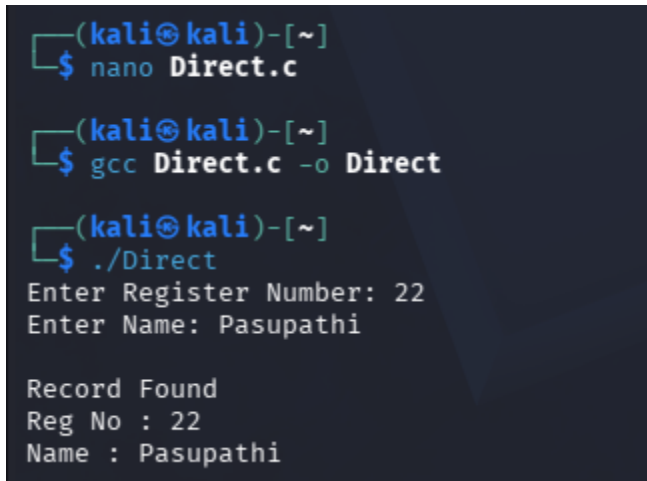
## PROGRAM B: DIRECT (RANDOM) FILE ORGANIZATION

### C PROGRAM

```
#include <stdio.h>  
  
struct student  
{  
    int regno;  
    char name[20];  
};  
  
int main()  
{  
    FILE *fp;  
    struct student s;  
    fp = fopen("random.dat", "wb+");  
    printf("Enter Register Number: ");  
    scanf("%d", &s.regno);  
    printf("Enter Name: ");  
    scanf("%s", s.name);  
    fwrite(&s, sizeof(s), 1, fp);  
    rewind(fp);  
    fread(&s, sizeof(s), 1, fp);  
    printf("\nRecord Found\n");  
    printf("Reg No : %d\n", s.regno);
```

```
printf("Name : %s\n", s.name);  
fclose(fp);  
return 0;  
}
```

Output:



```
(kali㉿kali)-[~]  
$ nano Direct.c  
  
(kali㉿kali)-[~]  
$ gcc Direct.c -o Direct  
  
(kali㉿kali)-[~]  
$ ./Direct  
Enter Register Number: 22  
Enter Name: Pasupathi  
  
Record Found  
Reg No : 22  
Name : Pasupathi
```

SHELL SCRIPT

```
#!/bin/bash  
  
echo "Enter Record:"  
  
read rec  
  
echo "$rec" > random.txt  
  
echo "Random Access Record"  
  
sed -n '1p' random.txt
```

Output:

```
(kali@kali)-[~]
$ nano lab13-2.sh

(kali@kali)-[~]
$ chmod +x lab13-2.sh

(kali@kali)-[~]
$ bash lab13-2.sh
Enter Record:
22
Random Access Record
22
```

## PROGRAM C: INDEXED FILE ORGANIZATION

### C PROGRAM

```
#include <stdio.h>

struct student
{
    int regno;
    char name[20];
};

int main()
{
    struct student s[3];
    int key, i;
    printf("Enter 3 Student Records\n");
    for(i=0;i<3;i++)
    {
        scanf("%d %s",&s[i].regno,s[i].name);
    }
    printf("Enter Register Number to Search: ");
    scanf("%d",&key);
```

```
for(i=0;i<3;i++)  
{  
if(s[i].regno==key)  
{  
printf("\nRecord Found\n");  
printf("Reg No : %d\n",s[i].regno);
```

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```
printf("Name : %s\n",s[i].name);  
return 0;  
}  
}  
printf("Record Not Found\n");  
return 0;  
}
```

Output:

```
(kali@kali)-[~]  
$ nano indexed.c  
  
(kali@kali)-[~]  
$ gcc indexed.c -o indexed  
  
(kali@kali)-[~]  
$ ./indexed  
Enter 3 Student Records  
22 Pasupathi  
23 Dhanush  
24 Gokul  
Enter Register Number to Search: 22  
  
Record Found  
Reg No : 22  
Name : Pasupathi
```

#### SHELL SCRIPT

```
#!/bin/bash  
  
echo "Enter Student Records"  
  
echo "101 Arun" > index.txt  
echo "102 Kumar" >> index.txt  
echo "103 Ravi" >> index.txt  
  
echo "Enter Register Number to Search:"  
  
read key  
  
grep "^$key" index.txt
```

Output:

```
(kali㉿kali)-[~]  
$ nano lab13-3.sh  
  
(kali㉿kali)-[~]  
$ chmod +x lab13-3.sh  
  
(kali㉿kali)-[~]  
$ bash lab13-3.sh  
Enter Student Records  
Enter Register Number to Search:  
22
```